

Accountability-by-Design OTA for AI-Enabled Safety-Critical Vehicle Functions

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Background

The automobile is now becoming software defined system, with much of its functionality being driven by software and artificial intelligence (AI). AI-enabled subsystems like Advanced driver assist systems (ADAS), perception systems, driver monitoring systems, and automated decision-making systems have a great role to play in assisting the vehicle operations that are very critical in safety [3][6]. These systems are based on sophisticated models of machine learning that constantly develop together with better performance, safety, and reliability of a manufacturer [3].

For sustaining and improving capabilities across the vehicle lifecycle, manufacturers rely more and more on Over-the-Air (OTA) software updates [1][2]. OTA technology allows delivering software patches, security updates, feature enhancements, and updates in AI models remotely through cloud infrastructure to vehicles [1]. Physical maintenance of the device is removed through this method and enables the device to update on a large scale [1][2].

The existing automotive OTA systems are mainly concerned with delivering secure and reliable software, applying tools such as authentication, encryption, secure boot, verification integrity, rollback protection, and version control [1][2]. Though mechanisms are able to block malicious updates, they typically view AI elements as usual software modules.

Besides, the recent OTA models mostly do not store AI-related provenance data such as model structure, training data, hyperparameters, model weights, and development history. To determine the precise AI system state that led to a specific behavior during safety-critical incidents is quite difficult[3][4]. In addition to that, new standards like the UNECE R156 focus on software traceability and accountability. Nevertheless, existing OTA architectures have no mechanisms to trace, explain, and reconstruct AI models, emphasizing the need to architecturally design and implement accountability-based OTA systems that are consistent with regulatory demands [1][2].

Research Hypothesis and Objectives

Research Hypothesis

Provenance of AI models, detailed evidence logging, and traceable deployment records can be integrated into existing secure OTA architectures to provide accountability-by-design in automotive OTA systems, which enhance traceability, explainability, and reconstructing incident capabilities of AI-enabled safety-critical vehicle functions and enable regulatory compliance [1][2][4].

Research Objectives

The study intends to develop and test an accountability-based OTA model that can enable traceability and responsibility accreditation of AI-based vehicle technologies.

The following are the specific aims of this research:

1. After defining existing architecture referenced to OTA in the automotive industry, examine the shortfalls of existing automotive OTA architectures to support AI model traceability and accountability [1][2].
2. Analyse regulatory and governance specifications in AI-enabled cars based on software updates, traceability, and attributing responsibility [3][6].
3. Establish the accountability and traceability criteria of deploying AI models in systems subjected to vehicle-safety [3][4].
4. Develop an accountability-by-design OTA architecture, including AI-based provenance tracking, version traceability, and organized evidence logging [1][4].

5. Provide a conceptual prototype or analysis scheme to prove and test the suggested architecture by showing what the proposed architecture can do in terms of providing explainability and incident reconstruction [4].

The eventual outcome of this research is to prevent the lack of a connection between secure OTA update systems and the responsibility mechanisms of AI-facilitated automotive systems, so that the implementation of AI can be considered more transparent and trustworthy in safety-critical vehicle contexts [1][3].

Methodology and programme

The proposed methodology is divided into five work packages (WPs) which explains how we plan to derive accountability-by-design requirements from regulations, do gap analysis, model the OTA lifecycle [7][9], and use a proof-of-concept OTA testbed and incident scenarios to evaluate traceability, explainability, and incident reconstruction, along with informing automotive practice and research guidelines [8]. It helps to address the difficulty of ensuring traceability along with accountability for safety-critical automotive systems in AI-enabled OTA updates.

WP1: Requirements along with Gap Analysis:

It identifies what ‘accountability-by-design’ should mean for OTA in AI-enabled vehicles and shortcomings of current solutions by providing us with the regulatory and conceptual foundation for the latter design and implementation work.

Task 1.1 Standards and Regulation review: Initially, we will collect and summarize essential standards and regulations (like ISO 24089, ISO 26262, ISO 21434, ISO 21448, ISO/PAS 8800, UN R156, UN R155, AI/AV liability, and AI governance papers) and obtain all details including OTA, logging, traceability, evidence, responsibility, and explainability [11][20].

Task 1.2 Accountability requirement extraction: By coding the clauses into requirement categories (traceability, explainability, evidence, roles/responsibilities, AI-specific aspects like model and data changes), we will compile an “accountability-by-design for OTA” requirements list [10].

Task 1.3 Review of current OTA solutions: Analysing the existing secure OTA architectures and industry practices (like security-by-design OTA, uptown style approaches, trusted execution environment-based OTA), we identify which features they provide (signing, encryption, rollback, basic logging).

Task 1.4 Analysis of gap: We will develop a matrix that aligns the identified accountability requirements against the capabilities of current OTA solutions, highlighting specific gaps, especially around AI-model level provenance, end-to-end traceability, and incident reconstruction.

WP2: Socio-Technical Modelling and Responsibility Mapping

This work package (WP2) addresses OTA as a socio-technical system, describing how technical flows, roles of organisation, and decision points interact. It ensures that the later architecture explicitly supports real responsibility chains, not just technical security.

Task 2.1: OTA lifecycle process modelling: We will select a representative AI-enabled safety-critical function (e.g., an ADAS component) and model the complete OTA lifecycle using BPMN/UML. It will include development, validation/verification, approval, safety/ cybersecurity assessment, campaign planning, deployment, monitoring, and recall/roll-back.

Task 2.2: Identification of Actor and Role: Next, we will determine who is involved at each stage in the process models by identifying all relevant actors (Tier1 supplier, OEM, OTA/cloud provider, regulator, fleet operator/driver, insurer) [13].

Task 2.3: Responsibility mapping (RACI): Later, we set the RACI (Responsible, Accountable, Consulted, Informed) matrix for major OTA decisions (like approving a model change, release of an update, issuing a

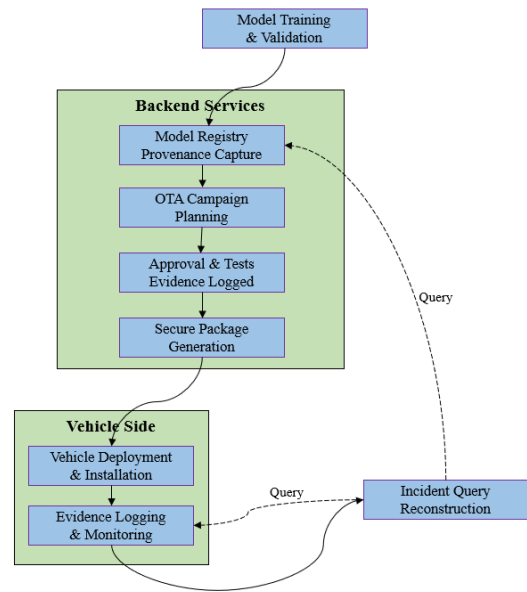


Figure : Accountability-oriented OTA architecture integrating AI model provenance registry, evidence logging, and incident reconstruction capabilities.

recall, and initiating rollback) and list important decision points requiring strong evidence and traceability [14].

Task 2.4: Experts' feedback/review and refinement (if feasible): Afterwards, we organize a small number of expert interviews or feedback sessions with engineers or researchers skilled in automotive OTA/AI to validate and refine the process and responsibility models.

WP3: Design of AI-Provenance and Accountability-Oriented OTA Architecture This work package, WP3, designs the data structures along with system architecture required to capture AI-specific provenance information and embed accountability-by-design into the OTA pipeline, building directly on the requirements and socio-technical models.

Task 3.1: Derivation of AI-provenance requirements: Based on WP1 and relevant AI-safety/explainable AI-literature, we ensure what must be tracked for AI models in vehicles: model identifiers, architecture, weight hashes, training data version/lineage, hyperparameters, validation metrics, deployment context (ECU/function, ODD assumptions) [18].

Task 3.2: Data and log schema design: Building concrete schemas (like JSON /relational tables) for AI_MODEL metadata, FUNCTION/CONFIGURATION, OTA_CAMPAIGN details, and EVIDENCE_LOG events, and we will specify how events are linked (e.g., via IDs and hash chaining) and protected (signatures) so that evidence is tamper-resistant [13].

Task 3.3: Accountability-oriented OTA architecture design: We will start off from a secure OTA reference (as identified in WP1), extend it with additional components such as a Model/Provenance Registry, an Evidence Logging service, and an Incident Reconstruction API/service [12][19]. Then draw high-level and component diagrams showing interactions between backend services and in-vehicle clients.

Task 3.4: Explainability and reconstruction hooks definition: Define how the stored provenance and logs will be used to answer typical investigation queries (like 'What model/configuration was deployed on vehicle X at time T?', 'Before deployment, which tests /approvals were completed?'), ensuring that these hooks directly support explainability and incident reconstruction.

WP4: Prototype Implementation

This work package translates the designed architecture into a working proof-of-concept OTA testbed, with both a baseline secure OTA pipeline and the proposed accountability-by-design extension.

Task 4.1: Testbed setup and AI function selection: We will use a simple OTA test environment consisting of a backend OTA server and a simulated vehicle ECU (e.g., container/VM). Next, we select a simple AI model that updates via OTA, representing a safety-related function[15].

Task 4.2: Baseline secure OTA implementation: Then we build a conventional security-by-design OTA pipeline: package creation, signing, verification, version checks, basic logging of update attempts and outcomes.

Task 4.3: Enhanced accountability-by-design implementation: The Model/Provenance Registry and Evidence Logging components will be designed and added in WP3 by us into OTA testbed. Extend the OTA flow to register AI models/configurations, capture approvals and test references, record deployments and rollbacks, and maintain tamper-resistant logs.

Task 4.4: Incident reconstruction interface: A simple dashboard/command-line interface will be used that allows querying by vehicle and time range to reconstruct software/AI model state, configuration, update history, and responsible approvals.

WP5: Scenario-Based Evaluation and Analysis

This work package (WP5) looks into the proposed architecture by using realistic incident scenarios on both the baseline and enhanced OTA setups, assessing improvements in traceability, explainability, and incident reconstruction capability.

Task 5.1: Scenario and metric design: After defining realistic OTA-related incident scenarios (e.g., update regression, partial deployment with rollback, recall verification) [16], we will formalise reconstruction tasks and metrics such as completeness of reconstruction, effort/time required, and clarity of responsibility trail [17].

Task 5.2: Experiment execution: We will execute the defined scenarios for each scenario on both the standard and proposed architectures, collecting logs and records, and reconstruction attempts.

Task 5.3: Comparative evaluation: We will execute reconstruction tasks using both architectures/measures defined metrics, and document where the proposed architecture enables faster or more complete reconstruction and clearer attribution of responsibility.

Task 5.4: Synthesis of findings and recommendations:

Thus, we will analyse the results, trade-offs (like overhead, implementation complexity), determine limitations, and derive practical design guidelines and future research directions for implementing accountability-by-design OTA in AI-enabled vehicles.

National importance

This research has national importance because modern vehicles are rapidly moving towards software-defined and AI-enabled systems. In systems like these, safety-critical functionalities are modified remotely through Over-the-Air (OTA) software updates. As these connected and automated vehicles are becoming more widespread, governments have introduced regulatory frameworks that treat software updates as a regulated component for the vehicle lifecycle. For example, the UNECE Regulation No. 156 requires automotive manufacturers to implement a Software Update Management System (SUMS) that makes sure that software updates are securely delivered, validated, and remain traceable throughout the lifetime of the vehicle by keeping a record of it [21]. Furthermore, ISO 24089 establishes engineering processes for safe and controlled vehicle software updates. It emphasises documentation, verification, and traceability after the vehicles' deployment [28]. These policies highlight the increasing need for reliable OTA architectures that are capable of supporting regulatory compliance and safety assurance.

However, the rapid integration of AI models in intelligent vehicle systems, especially for decision-making, makes way for new challenges related to transparency and accountability. Machine-learning(ML) models often operate as opaque “black boxes,” which makes it difficult to retrace system behaviour during safety-critical incidents or determine which model configuration was active at the time of failure [24]. Addressing this issue has been a priority for the UK AI industry and the UK automotive sector, both of which are recognised as strategic priorities in the UK National AI Strategy [27]. Developing accountability-by-design OTA architectures would support safer deployment of AI-enabled vehicles, strengthen regulatory oversight, and improve public trust in autonomous systems. Such frameworks can help reduce not only the costly recalls and liability risks, but the industry can also enable UK automotive manufacturers and technology companies to deploy AI updates more safely and efficiently [23].

Academic relevance and impact

The academic relevance of this project can be associated with its contribution to transparency in accountability and transparency in AI within OTA systems in AI-enabled vehicles in safety-critical applications. With the increased use of software-defined features in vehicles and the use of artificial intelligence in these systems, OTA updates have become essential in enhancing the overall performance and safety of these vehicles. However, while OTA systems have been effective in ensuring the safe delivery of updates to vehicles, they have been limited in providing detailed evidence regarding the specific AI used in the updates [36][37][41].

One important contribution that this research aims to make is to the field of automotive OTA system security and cybersecurity. Current research is focused on secure updates, software integrity, authenticity, and prevention of malicious interference [40][31]. Although all these areas are extremely critical and important, they are only prevention-centric and lack sufficient support for reconstruction in case there is an occurrence of security breach. This research aims to contribute to existing research by focusing on accountable OTA systems rather than secure updates [39][31].

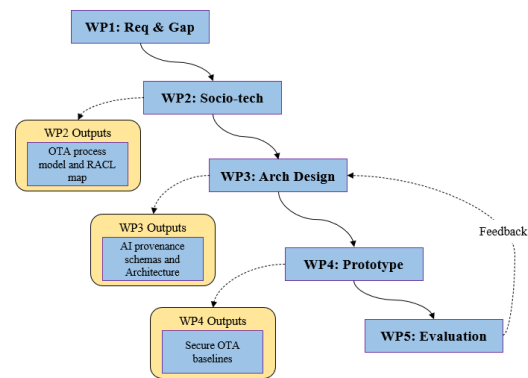


Figure : Relationship between WPs

The project also contributes to the field of AI accountability and governance by considering it an architectural aspect rather than an ethical or legal aspect. This research also provides an understanding of how accountability can be achieved with respect to AI models, including aspects such as version, data used for training, hyperparameters, weights, and history, which gives an understanding of how it can be achieved in a real-world vehicle system, considering that it directly impacts human safety in safety-critical scenarios [41][35].

Additionally, the research would also contribute to the domains of explainable AI, trustworthy AI, and socio-technical systems. The research would also extend the domain of explainable AI not only to include model decision interpretation but also to include model deployment history and incident reconstruction. The research would also understand that accountability in OTA is not only dependent on technology but also on organization [31].

Overall, this project contributes to the concept of accountability-by-design by expanding existing security-by-design principles with provenance tracking, logging, and evidence generation. The academic contribution of this project is that it would set the stage for future research in the domain of trustworthy AI, autonomous vehicles, digital forensics, software traceability, and regulatory assurance in safety-critical systems [36][35].

Risk management

Developing an OTA system for accountability has challenges in all five work packages. Prototype implementation (WP4) has a major risk of technical complexity. The implementation of the prototype will become slow if we push it too far. Secure OTA baseline from Iyieke et al. [42] will address this issue with proven statements, and we will be able to add AI provenance for logging. We will provide a focused query interface for traceability, in case full incident reconstruction becomes difficult.

Modelling in WP2 and validating in WP5 is another concern with respect to accessing the experts. The published TARA case studies can be used for academic contacts, and in the worst case, self-evaluation can be made against the standard OTA failure modes and metrics of AI accountability, which ensure that results remain sturdy.

We need to be careful on the updates provided on stable foundations of UN R156 and ISO 26262 in order to stick with the regulatory changes happened during the project. When it comes to WP3, the architecture design stays on track by adapting to the existing AI model schemas instead of building a new one from the scratch. For the evaluation in WP5, we will use CARLA and open AV datasets to overcome the limitations with simulators. All these mitigation plans ensure that the project delivers the solid research activity on automotive practice.

Work Plan

[illegible]

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Group Member Contributions

Group member name	Main contributions and other work carried out collectively with other group members.
Rushbah Awan	National Importance, Gantt chart , Report writing
Anil Emanuel	Academic Relevance and Impact,Report writing
Ashrith Reddy Danagari	Risk Management and Master Diagram
Simanta Mostafa	Methodology and Task identification ,Report writing
Deepakraj Rajmohan	Ideation, Programme and Report writing
Sayed Omar Aabid	Background and Introduction, Report writing

Assessment and Feedback Form

This is to be completed by the marker and not the group and will be used to provide feedback. Therefore this document when submitted **must not be as a pdf file and submitted in Word format** so the below form can be completed.

Assessment component	Mark
<i>Vision of proposed work</i>	
- is of excellent quality and importance within or beyond the field(s) or area(s)	/20
- has the potential to advance current understanding, generates new knowledge, thinking or discovery within or beyond the field or area	/10
- is timely given current trends, context and needs	/10
<i>The approach to deliver the work</i>	
- is effective and appropriate to achieve your objectives	/15
- summarises previous work and describes how it will be built upon and progressed	/10
- will maximise research outputs and impacts	/10
<i>The work plan</i>	
- provides a clear, realistic, and well-structured timeline for the project	/5
- includes key milestones, tasks, and deliverables that align with the research objectives	/5
<i>Quality of proposal writing</i>	
- Quality of writing in clear level of English	/5
- Neat presentation of the proposal including figures	/5
- Correct referencing done as required	/5
TOTAL	/100

What has been done well in the proposal:

What could have been done better in the proposal:

General comments: